

Brandon Viaje

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EDUCATION

Ontario Tech University

Oshawa, ON

Bachelor of Science in Computer Science Co-op

Apr 2027

- GPA: 4.00/4.30 (President's List x4)
- Relevant Coursework: Compilers, Database Systems, Operating Systems, Data Structures & Algorithms, Machine Learning, Systems Programming

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript, SQL

Systems & Low-Level: CUDA, Compiler Design, Concurrency, Multithreading, Memory Management, Linux/Unix

Libraries & Frameworks: NumPy, PyTorch, Spring Boot, React, Node.js

Tools & Infrastructure: CMake, Docker, Git, Google Cloud (GCP), CI/CD, ONNX

Databases: MongoDB, PostgreSQL, MySQL

PROJECTS

Compiler | C++, CMake, GitHub Actions

Oct. 2025 – Jan. 2026

- Developing a custom compiler using a Recursive Descent Parser and an LL(1) grammar
- Architected a high-performance Lexer using `std::string_view` for zero-copy string manipulation
- Designed Abstract Syntax Tree using `std::unique_ptr` for safe, automated memory management of program nodes
- Integrated a CI/CD pipeline with GitHub Actions and Clang-Tidy to enforce strict static analysis and memory safety standards

Chess Engine | C, Google Cloud, Lichess API

Oct. 2025 – Jan. 2026

- Engineered a UCI chess engine in C using Bitboards, achieving 1.2M positions/sec on a single thread
- Optimized search efficiency by implementing Alpha-Beta Pruning and Negamax to increase decision depth
- Deployed to Google Cloud Compute Engine for continuous automated matchmaking via Lichess API
- Implemented transposition tables to cache board states, eliminating redundant calculations
- Stress-tested engine across 100+ live matches via Lichess API to validate runtime stability under continuous play

Deep Learning Inference Engine | C++, ONNX, Google Protobuf

Nov. 2025 – Present

- Built a C++ runtime to execute ML models efficiently by reconstructing computation graphs
- Implemented topological sort to order graph nodes, ensuring all dependencies are resolved before execution
- Utilized Google Protobuf to deserialize model weights and metadata into a custom internal representation
- Designed an extensible Operator Registry, allowing for decoupled integration of new layer types

Voice Assistant Discord Bot | Python, Asyncio, OpenAI Whisper, NumPy

Dec. 2025 – Feb. 2026

- Built a concurrent audio pipeline using Asyncio, handling multiple voice streams with sub-second latency
- Integrated wake-word detection to process audio triggers on-device, optimizing cloud API usage and token costs.
- Implemented packet aggregation to reassemble fragmented audio streams and resolve network jitter
- Designed role-based permissions to secure command execution across shared user channels

Movie Discovery Platform | Docker, Spring Boot, PostgreSQL, JavaScript, Playwright

Jan. 2025 – Apr. 2025

- Led a team of 5 in an Agile/Scrum environment, managing sprint backlogs and conducting daily stand-ups to ensure on-time delivery
- Built frontend in JavaScript (ES6+), optimizing DOM manipulation and asynchronous API calls
- Developed a RESTful API with Spring Boot, optimizing PostgreSQL database queries to reduce response times
- Achieved 80% code coverage using JUnit integration tests to catch bugs before deployment